

HBSC

Updated: 2026-09-04

Data Overview

It's a huge file:

```
## [1] 244097    120
```

With these columns:

HBSC	seqno_int	cluster	countryno	region
id1	id2	id3	id4	weight
adm	month	year	age	agecat
sex	grade	monthbirth	yearbirth	fasfamcar
fasbedroom	fascomputers	fasbathroom	fasdishwash	fasholidays
health	lifesat	headache	stomachache	backache
feellow	irritable	nervous	sleepdifficulty	dizzy
thinkbody	physact60	breakfastwd	breakfastwe	fruits_2
vegetables_2	sweets_2	softdrinks_2	fmeal	toothbr
timeexe	smokltm	smok30d_2	alc1tm	alc30d_2
drunkltm	drunk30d	cannabis1tm_2	cannabis30d_2	bodyweight
bodyheight	likeschool	schoolpressure	studtogether	studhelpful
studaccept	teacheraccept	teachercare	teachertrust	bulliedothers
beenbullied	cbulliedothers	cbeenbullied	fight12m	injured12m
friendhelp	friendcounton	friendshare	friendtalk	emconlfreq1
emconlfreq2	emconlfreq3	emconlfreq4	emconlpref1	emconlpref2
emconlpref3	emcsocmed1	emcsocmed2	emcsocmed3	emcsocmed4
emcsocmed5	emcsocmed6	emcsocmed7	emcsocmed8	emcsocmed9
hadsex	agesex	contraceptcondom	contraceptpill	countryborn
countrybornmo	countrybornfa	motherhome1	fatherhome1	stepmohome1
stepfahome1	fosterhome1	elsehome1_2	employfa	employmo
employnotfa	employnotmo	talkfather	talkstepfa	talkmother
talkstepmo	famhelp	famsup	famtalk	famdec
MBMI	IRFAS	IRRELFAS_LMH	IOTF4	oweight_who

So after selecting and grouping variables, create dat with :

```
## ## Individual Information
## * 'seqno_int'
## * 'countryno'
## * 'weight'
## * 'agecat'
## * 'sex'
## * 'MBMI'
```

```

## * 'IRFAS'
## * 'IRRELFAS_LMH'
## * 'oweight_who'
## * 'bodyweight'
## * 'bodyheight'
##
## ## Health Issues and Life Satisfaction
## * 'health'
## * 'lifesat'
## * 'headache'
## * 'stomachache'
## * 'backache'
## * 'feellow'
## * 'irritable'
## * 'nervous'
## * 'sleepdifficulty'
## * 'dizzy'
##
## ## Problematic Social Media Use (PMSU)
## * 'emcsocmed1'
## * 'emcsocmed2'
## * 'emcsocmed3'
## * 'emcsocmed4'
## * 'emcsocmed5'
## * 'emcsocmed6'
## * 'emcsocmed7'
## * 'emcsocmed8'
## * 'emcsocmed9'
##
## ## Bullying
## * 'beenbullied'
## * 'cbeenbullied'
##
## ## Communicates Online More Than Real Life
## * 'emconlfreq1'
## * 'emconlfreq2'
## * 'emconlfreq3'
## * 'emconlfreq4'
## * 'emconlpref1'
## * 'emconlpref2'
## * 'emconlpref3'
##
## ## Family Support
## * 'famhelp'
## * 'famsup'
## * 'famtalk'
## * 'famdec'
##
## ## Friends Support
## * 'friendhelp'
## * 'friendcounton'
## * 'friendshare'
## * 'friendtalk'
##

```

```

## ## School Support
## * 'likeschool'
## * 'schoolpressure'
## * 'studtogether'
## * 'studhelpful'
## * 'studaccept'
## * 'teacheraccept'
## * 'teachercare'
## * 'teachertrust'
##
## ## Communication with Parents
## * 'talkfather'
## * 'talkmother'
##
## ## Good/Bad Habits
## * 'physact60'
## * 'timeexe'
## * 'alc30d_2'

```

The number of missing values is relevant and need to address later:

##	variable	na_count	na_pct
## 1	oweight_who	54573	22%
## 2	MBMI	53311	22%
## 3	bodyweight	41128	17%
## 4	bodyheight	40036	16%
## 5	emcsocmed8	32878	13%
## 6	emcsocmed9	32855	13%
## 7	emcsocmed7	32597	13%
## 8	emcsocmed6	32544	13%
## 9	emcsocmed5	32443	13%
## 10	emcsocmed4	32428	13%
## 11	emcsocmed3	32009	13%
## 12	emcsocmed1	31973	13%
## 13	emcsocmed2	31659	13%
## 14	emconlpref3	31097	13%
## 15	emconlpref2	30586	13%
## 16	emconlpref1	29813	12%
## 17	emconlfreq2	29552	12%
## 18	emconlfreq4	29003	12%
## 19	emconlfreq3	28973	12%
## 20	emconlfreq1	26720	11%
## 21	famdec	20237	8%
## 22	famsup	20091	8%
## 23	famtalk	20019	8%
## 24	alc30d_2	19326	8%
## 25	famhelp	18896	8%
## 26	teachercare	18340	8%
## 27	teachertrust	17839	7%
## 28	talkfather	17491	7%
## 29	studaccept	17444	7%
## 30	beenbullied	16962	7%
## 31	teacheraccept	16764	7%
## 32	studhelpful	16746	7%

```
## 33 studtogether 16351 7%
## 34 talkmother 15521 6%
## 35 cbeenbullied 15084 6%
## 36 IRFAS 12599 5%
## 37 IRRELFAS_LMH 12599 5%
## 38 sleepdifficulty 10814 4%
## 39 friendshare 10813 4%
## 40 friendtalk 10808 4%
## 41 friendcounton 10374 4%
## 42 friendhelp 9744 4%
## 43 dizzy 8566 4%
## 44 fellow 6696 3%
## 45 backache 6543 3%
## 46 irritable 6539 3%
## 47 nervous 6525 3%
## 48 physact60 6215 3%
## 49 stomachache 5679 2%
## 50 schoolpressure 5486 2%
## 51 likeschool 5109 2%
## 52 timeexe 4799 2%
## 53 lifesat 4512 2%
## 54 headache 4505 2%
## 55 health 2577 1%
## 56 agecat 1516 1%
## 57 seqno_int 0 0%
## 58 countryno 0 0%
## 59 weight 0 0%
## 60 sex 0 0%
```

Now I explore all variables by group and feature engineer as appropriate.

[1] “Individual Information”

11 vars here. So do 1x1.

```
## [1] "seqno_int" "countryno" "weight" "agecat" "sex"
## [6] "MBMI" "IRFAS" "IRRELFAS_LMH" "oweight_who" "bodyweight"
## [11] "bodyheight"
```

seqno_int is just an identifier

countryno is recoded to country_name and this is the distribution.

Table 2: #/% of observations by country

country_name	n	pct
Albania	1765	1%
Armenia	4717	2%
Austria	4129	2%
Azerbaijan	4586	2%
Belgium (Flemish)	4333	2%

country_name	n	pct
Belgium (French)	5578	2%
Canada	12950	5%
Croatia	5169	2%
Czech Republic	11564	5%
Denmark	3181	1%
England	3397	1%
Estonia	4725	2%
Finland	3146	1%
France	9170	4%
Georgia	4242	2%
Germany	4347	2%
Greenland	1243	1%
Hungary	3789	2%
Iceland	6996	3%
Ireland	3833	2%
Israel	7712	3%
Italy	4144	2%
Kazakhstan	4868	2%
Latvia	4412	2%
Lithuania	3797	2%
Luxembourg	4070	2%
Macedonia	4658	2%
Malta	2576	1%
Netherlands	4698	2%
Norway	3127	1%
Poland	5224	2%
Portugal	6126	3%
Republic of Moldova	4686	2%
Romania	4567	2%
Russia	4281	2%
Scotland	5021	2%
Serbia	3933	2%
Slovakia	4785	2%
Slovenia	5667	2%
Spain	4320	2%
Sweden	4185	2%
Switzerland	7510	3%
Turkey	5848	2%
Ukraine	6660	3%
Wales	15951	7%
NA	8411	3%

weight is unrelated so ignore/remove TBC?

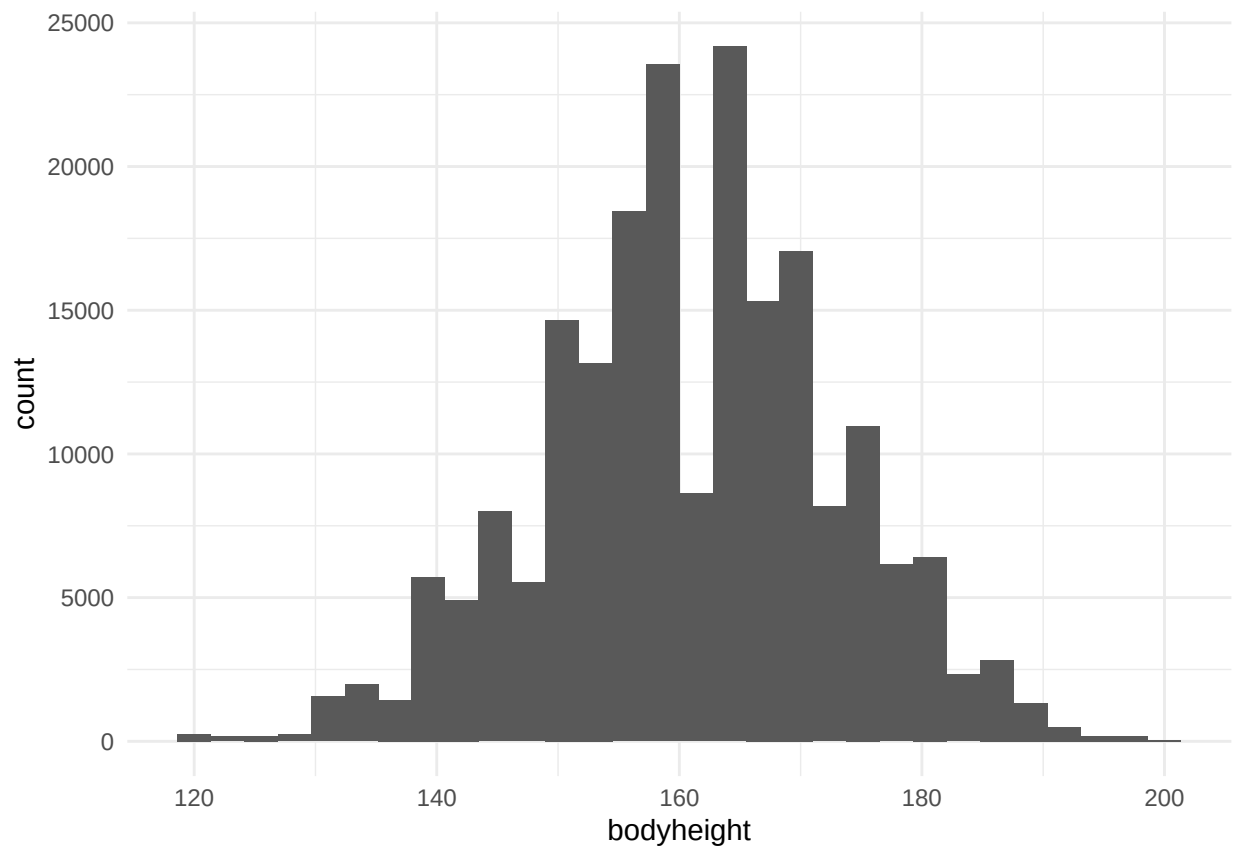
bodyheight seems to be cm and range from 120 to 200 so will use. Impute NAs?

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##    120.0   153.0   161.0   161.1   170.0   200.0   40036
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```

```
## Warning: Removed 40036 rows containing non-finite outside the scale range
```

```
## ('stat_bin()').
```

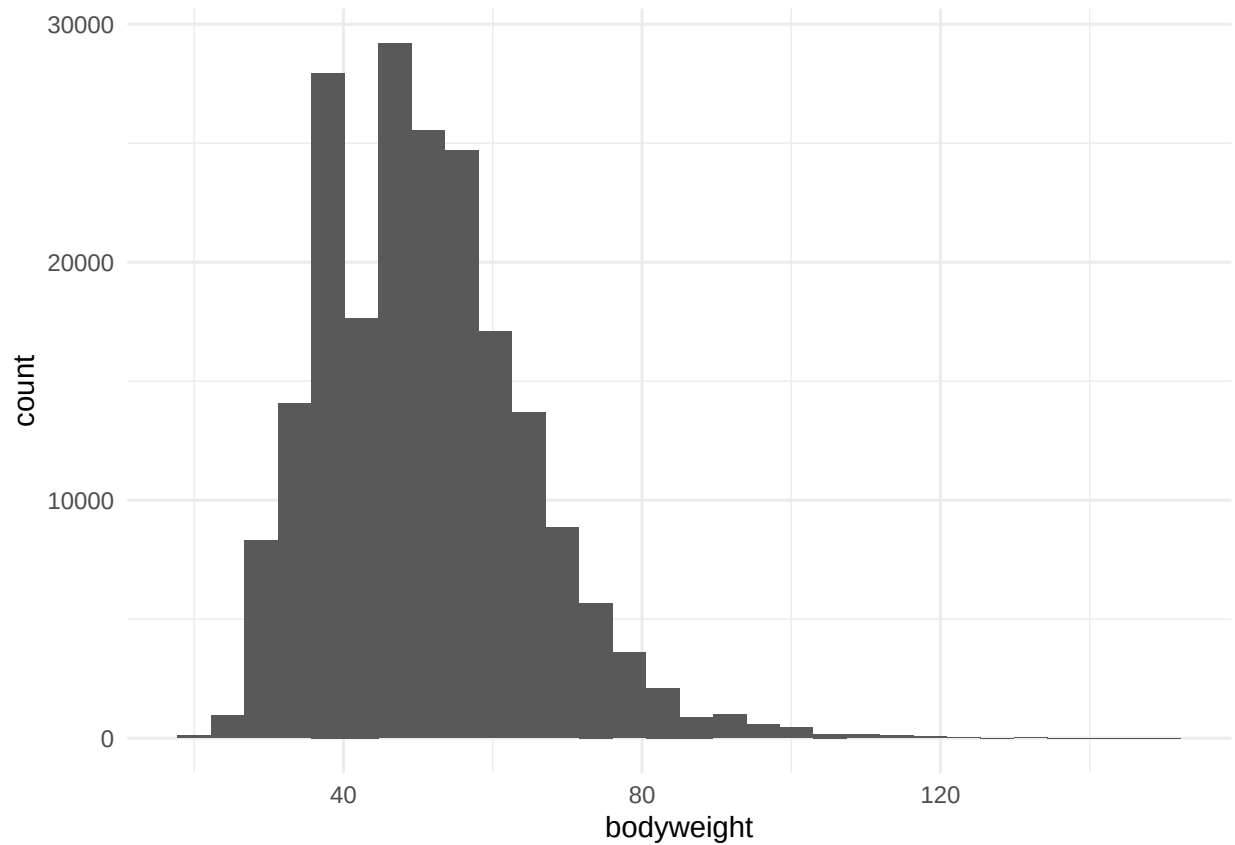


bodyweight seems to be kg? and range from 20 to 150 so will use maybe. Impute NAs?

```
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's  
##  20.00  40.00   50.00   51.07  60.00  150.00  41128
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```

```
## Warning: Removed 41128 rows containing non-finite outside the scale range  
## ('stat_bin()').
```

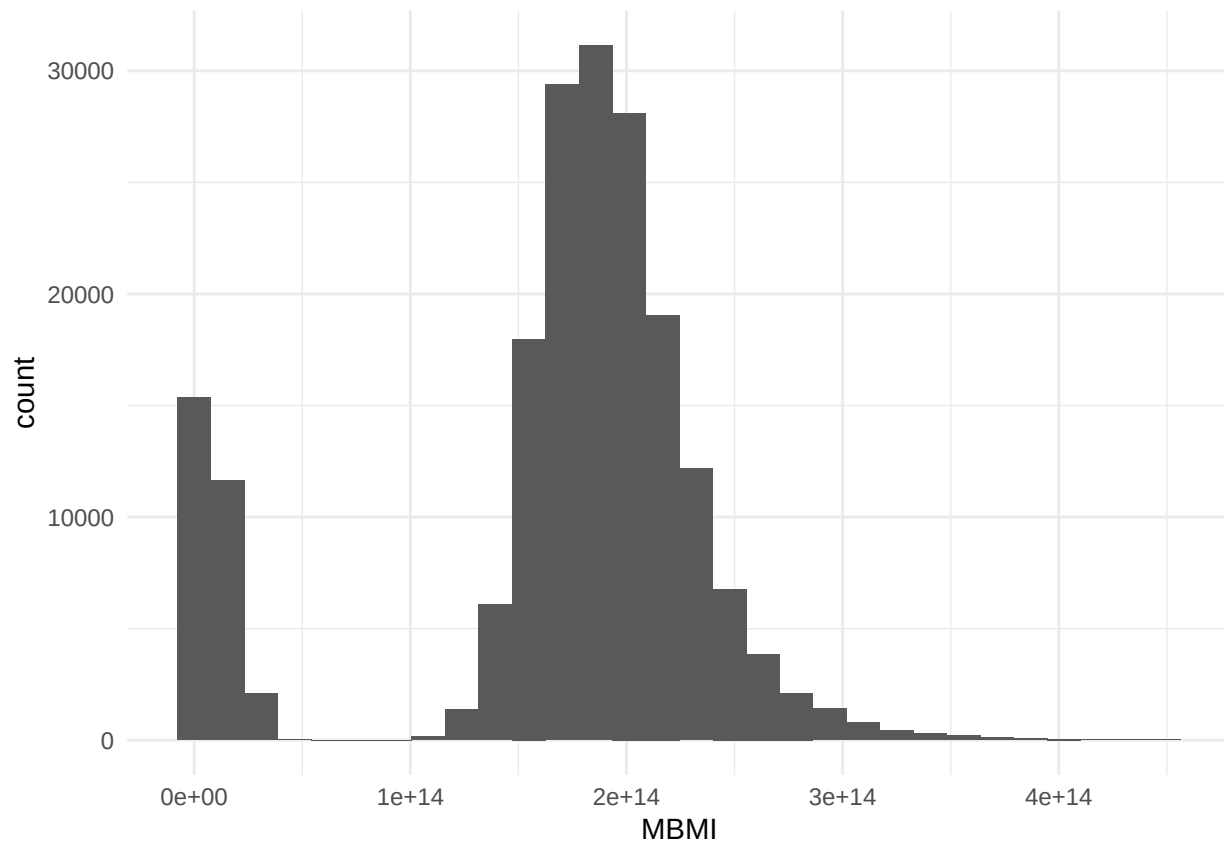


MBMI TBC huge odd range of #s.

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.      NA's
## 0.000e+00 1.579e+14 1.837e+14 1.671e+14 2.086e+14 4.489e+14   53311
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```

```
## Warning: Removed 53311 rows containing non-finite outside the scale range
## ('stat_bin()').
```



oweight_who seems to be binary??? TBC. Not sure if helpful to add.

```
##
##      0      1   <NA>
## 150874 38650 54573
```

agecat has 3 values and will leave as is: 1 ~ 11, 2 ~ 13, 3 ~ 15.

```
##
##      1      2      3 <NA>
## 81834 83749 76998 1516
```

sex is binary with 1 (boy), 2 (girl) so will leave as is.

```
##
##      1      2   <NA>
## 120350 123747      0
```

IRRELFAS_LMH is relative family affluence with 1 Low20, 2 Mid60, 3 High20. Leave as is.

IRFAS is discrete [0,13]. TBC what it means and rel with LMH cause it's not mut exclusive.

```
##
##      1      2      3 <NA>
##      0      715      0      0      0
```

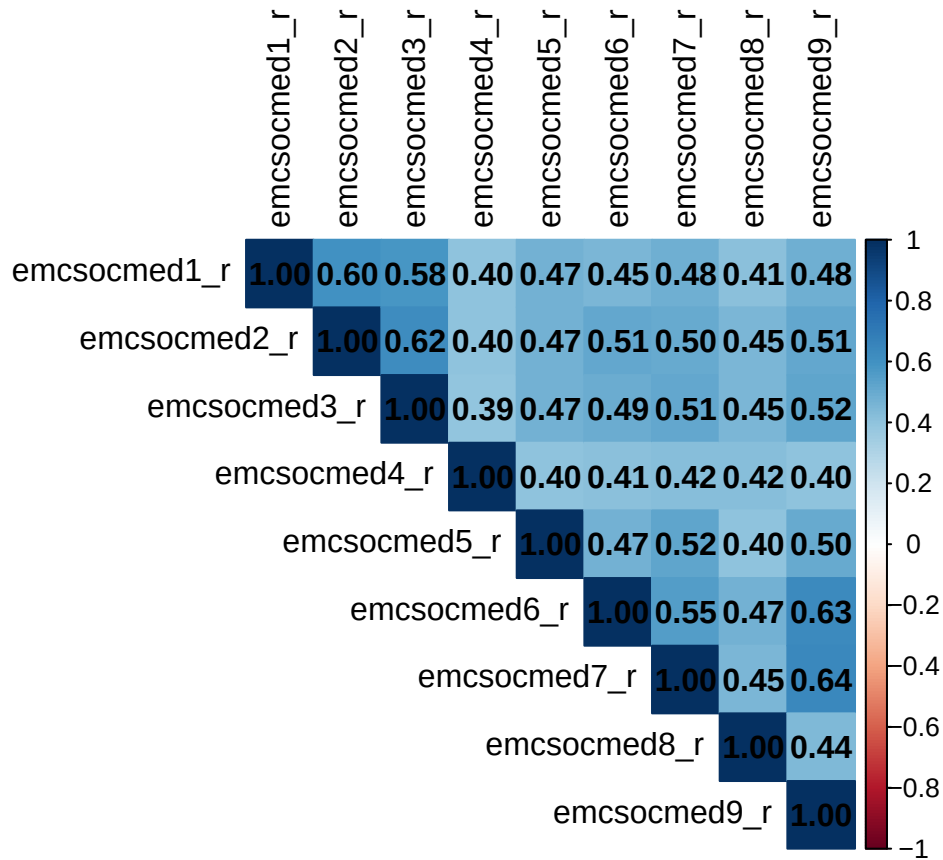
```
## 1 2238 2 0 0
## 2 4780 270 0 0
## 3 5217 3609 0 0
## 4 6829 6272 0 0
## 5 8199 9442 0 0
## 6 8522 13305 331 0
## 7 7593 17420 1513 0
## 8 1724 26873 2120 0
## 9 1 29479 2672 0
## 10 0 23830 5408 0
## 11 0 11158 11359 0
## 12 0 488 13893 0
## 13 0 0 6236 0
## <NA> 0 0 0 12599
```

[3] “Problematic Social Media Use (PMSU)”

9 binary variables are recoded to 0/1.

item	topic
emcsocmed1	Social media: I can't think of anything else
emcsocmed2	Social media: Dissatisfied cause want to spend more time
emcsocmed3	Social media: Felt bad cause want to play
emcsocmed4	Social media: Failed to spend less time
emcsocmed5	Social media: Neglected other activities
emcsocmed6	Social media: Had arguments because of use
emcsocmed7	Social media: Lied about amount of time spent
emcsocmed8	Social media: Use it to escape from negative feelings
emcsocmed9	Social media: Conflict with family because of use

And they are correlated.



These 9 vars are sumed and then binned to H/M/L where H(6-9), M(2-5) and L(0-1). Resulting in:

	1	2	3	NA
0	72474	0	0	0
1	36277	0	0	0
2	0	26066	0	0
3	0	19012	0	0
4	0	14908	0	0
5	0	10306	0	0
6	0	0	5905	0
7	0	0	3382	0
8	0	0	1878	0
9	0	0	2979	0
NA	0	0	0	50910

[4] “Bullying”

There are 2 variables that measure if kid has been bullied. a) beenbullied: Been bullied past months and b) cbeenbullied: Been cyber bullied. Answers range from 1 (Haven’t) to 5 (Several times a week).

I will derive 2 to keep: cbeenbullied as y/n and bulliedsum as sum of both.

```
## cbullied recoding:
```

```
##
##           0           1    <NA>
##  1    198900           0           0
##  2           0    20426           0
##  3           0     4625           0
##  4           0     2274           0
##  5           0     2788           0
##  <NA>           0           0    15084
```

bullying score: (low is better)

score	n	pct
2	151013	62%
3	34034	14%
4	13753	6%
5	7360	3%
6	7703	3%
7	2745	1%
8	1288	1%
9	747	0%
10	1093	0%
NA	24361	10%

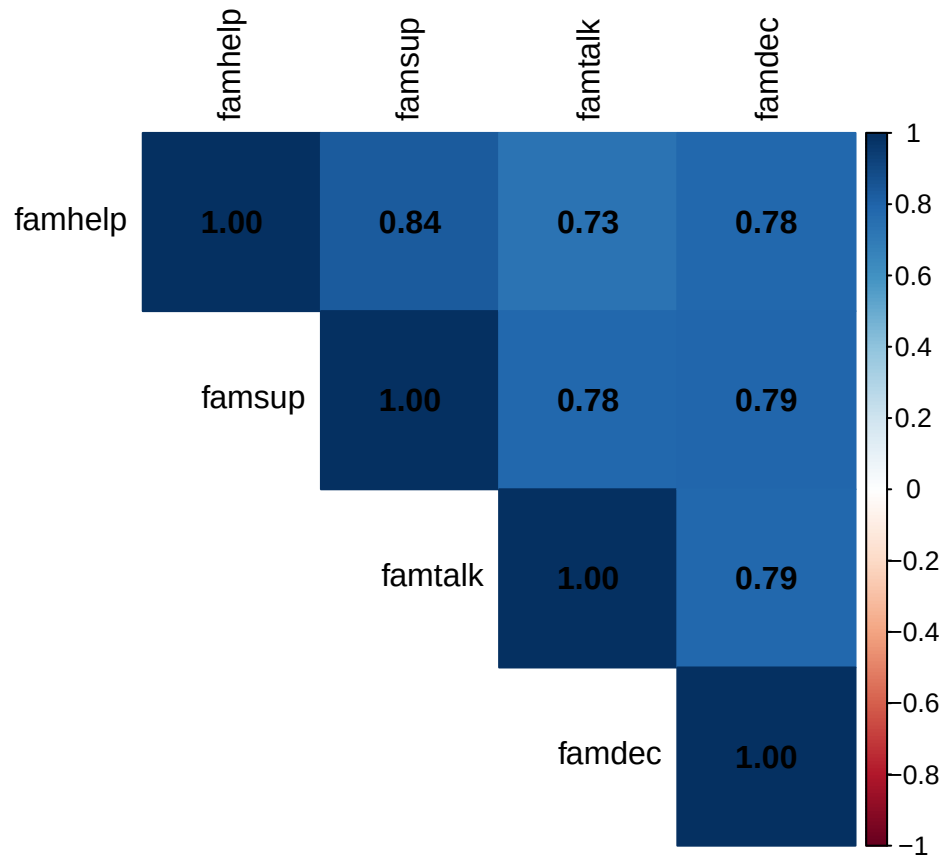
[6] “Family Support”

4 questions going from 1 Very strongly disagree to 7 Very strongly agree.

```
## [1] "famhelp" "famsup" "famtalk" "famdec"
```

item	topic
famhelp	Family tries to help
famsup	Get emotional help from family
famtalk	Talk about problems with family
famclose	Feel close to family

And they are correlated.



Summing them results in [4,28] and then grouped as H/M/L as L 4-11, M 12-19, H 20-28. Resulting in:

	1	2	3	NA
4	10939	0	0	0
5	1597	0	0	0
6	1452	0	0	0
7	1382	0	0	0
8	2477	0	0	0
9	1485	0	0	0
10	2274	0	0	0
11	1739	0	0	0
12	0	2655	0	0
13	0	2279	0	0
14	0	2745	0	0
15	0	2781	0	0
16	0	5747	0	0
17	0	3440	0	0
18	0	4455	0	0
19	0	4637	0	0
20	0	0	7739	0
21	0	0	5853	0
22	0	0	9240	0
23	0	0	7878	0
24	0	0	13474	0
25	0	0	11089	0

	1	2	3	NA
26	0	0	13090	0
27	0	0	15938	0
28	0	0	85359	0
NA	0	0	0	22353

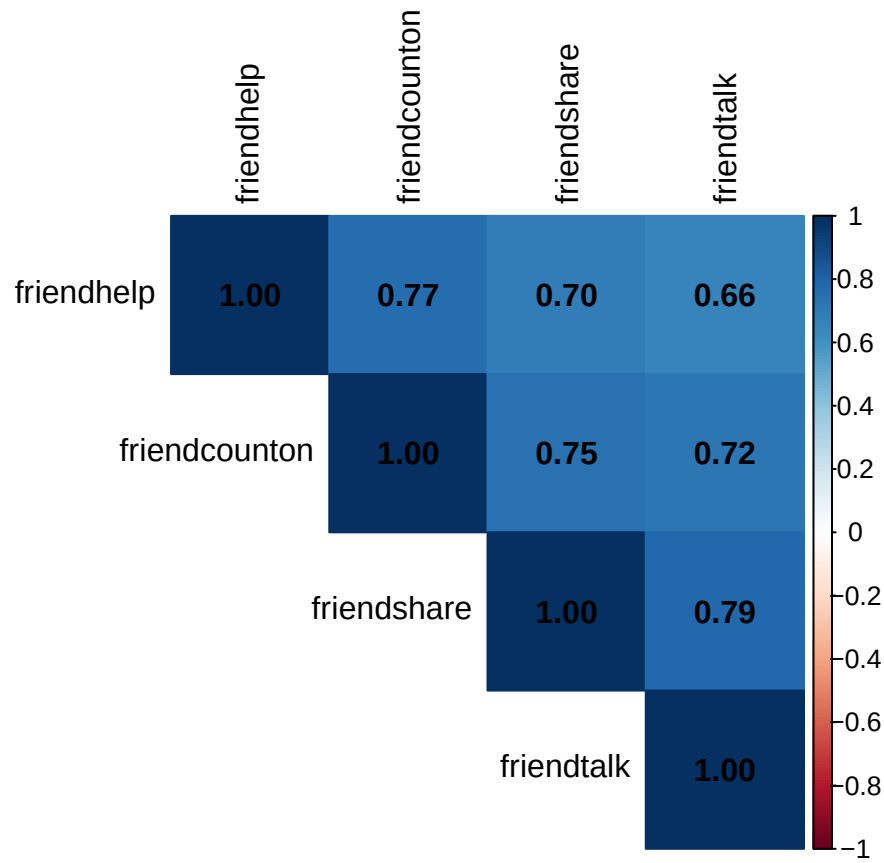
[7] “Friends Support”

4 questions going from 1 Very strongly disagree to 7 Very strongly agree.

```
## [1] "friendhelp"      "friendcounton" "friendshare"    "friendtalk"
```

item	topic
friendhelp	My friends really try to help me
friendcounton	I can count on my friends when things go wrong
friendshare	I have friends with whom I can share my joys and sorrows
friendtalk	I can talk about my problems with my friends

And they are correlated.



Summing them results in [4,28] and then grouped as H/M/L as L 4-11, M 12-19, H 20-28. Resulting in:

	1	2	3	NA
4	9995	0	0	0
5	2159	0	0	0
6	2109	0	0	0
7	2392	0	0	0
8	3798	0	0	0
9	2463	0	0	0
10	3752	0	0	0
11	2869	0	0	0
12	0	4267	0	0
13	0	3834	0	0
14	0	4428	0	0
15	0	4339	0	0
16	0	9075	0	0
17	0	5750	0	0
18	0	7126	0	0
19	0	7644	0	0
20	0	0	11463	0
21	0	0	8358	0
22	0	0	12583	0
23	0	0	10381	0
24	0	0	16855	0
25	0	0	12468	0
26	0	0	12977	0
27	0	0	14317	0
28	0	0	55799	0
NA	0	0	0	12896

[9] “Communication with Parents”

Two vars with answers from 1 (Very easy) to 5 (Don’t have or see). With this counts and proportions in data.

```
## [1] "talkfather" "talkmother"
```

```
##           talkmother
## talkfather    1      2      3      4      5 <NA>
##      1  62315 6269 2345  842  891 1115
##      2  31505 40099 4150  959  753  880
##      3   9936 17764 10530 1130  499  430
##      4   3719  5070  4602 5132  294  193
##      5   5842  4188  1968 1039 1905  242
##      <NA> 2582  1523   447  174  104 12661
```

```
##           talkmother
## talkfather    1      2      3      4      5 <NA>
##      1  0.255 0.026 0.010 0.003 0.004 0.005
##      2  0.129 0.164 0.017 0.004 0.003 0.004
##      3  0.041 0.073 0.043 0.005 0.002 0.002
##      4  0.015 0.021 0.019 0.021 0.001 0.001
##      5  0.024 0.017 0.008 0.004 0.008 0.001
##      <NA> 0.011 0.006 0.002 0.001 0.000 0.052
```

With some correlation between them (I expected more).

So I will dichotomize like HBSC and keep sum of both as well for SHAP. These are dichotomized for each parent as Yes if [1,2] and No if [3-5].

Where new vars look like this for dichotom values:

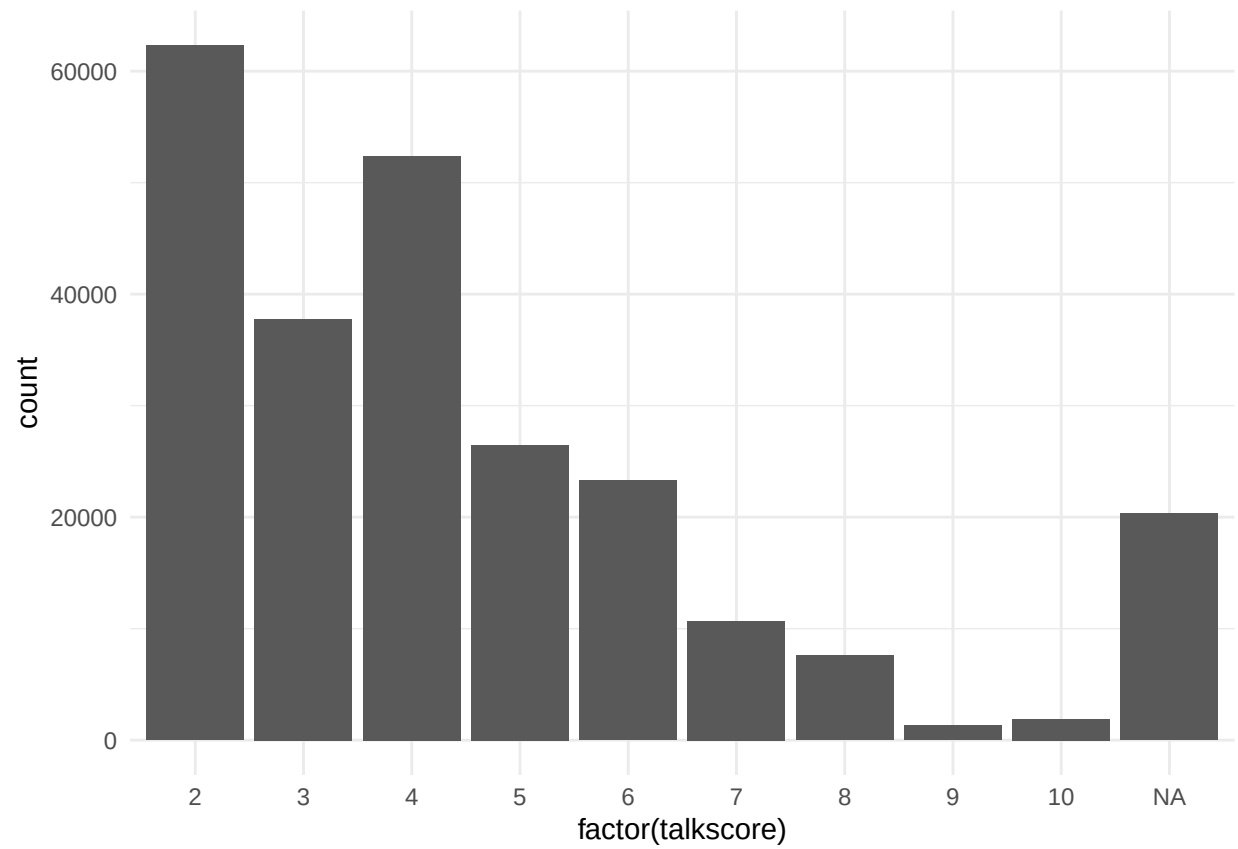
Father:

```
##
##           0      1  <NA>
##  1           0 73777      0
##  2           0 78346      0
##  3    40289      0      0
##  4    19010      0      0
##  5    15184      0      0
##  <NA>      0      0 17491
```

Mother:

```
##
##           0      1  <NA>
##  1           0 73777      0
##  2           0 78346      0
##  3    40289      0      0
##  4    19010      0      0
##  5    15184      0      0
##  <NA>      0      0 17491
```

And the talk score looks like this. Will keep for now and orig vars as well.



[2] "Health Issues and Life Satisfaction"
[5] "Communicates Online More Than Real Life" [8] "School Support"
[10] "Good/Bad Habits"